॥वसुधैव कुटुम्बकम्॥

# SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

**Symbiosis International (Deemed University)**

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

## Assignment No. 03

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|---|---|
| **Subject:** | DevOps Lab |
| **TCODE:** | TE7950 |
| **Name of Student** | **Raghav Sharma** |
| **PRN No.** | **23070122171** |
| **Branch** | CSE, Batch (2023-27), L3 |
| **Academic Year & Semester** | 25 – 26, Sem VII |
| **Date of Performance** | 23rd July, 2026 |
| **Title of Assignment:** | Simulating Git Collaboration & Conflict Resolution |
| **Theory:** | Git collaboration enables multiple developers to work on the same project using remote repositories. Merge conflicts occur when different users modify the same portion of a file. Git provides mechanisms to identify, resolve, and merge conflicting changes while preserving the project's version history. |
| **Implementation:** | Step 1: Create a new branch named conflict and verify the available branches using the git branch command.<br>`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git branch conflict`<br><br>`C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git branch`<br>`  conflict`<br>`  feature`<br>`* main`<br><br>Step 2: Make changes in the main branch and the conflict branch, stage the changes, and commit them using the git add and git commit commands. |

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git add .

C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git commit -m "Making Conflict"
[main c57ec4d] Making Conflict
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git checkout conflict
Switched to branch 'conflict'

C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git add .

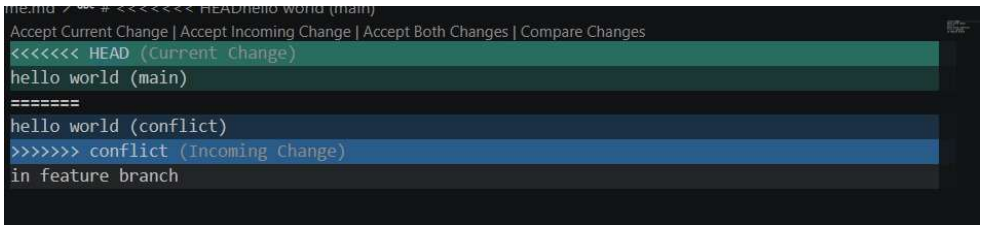
C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git commit -m "On Conflict"
[conflict 139abab] On Conflict
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Step 3: Switch back to the main branch and merge the conflict branch using the git merge conflict command.

```
C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git checkout main
Switched to branch 'main'

C:\Users\sharm\Desktop\7th sem\DevOpsL\devopsL\Github>git merge conflict
Auto-merging readme.md
CONFLICT (content): Merge conflict in readme.md
Automatic merge failed; fix conflicts and then commit the result.
```

Step 4: Observe the merge conflict message, resolve the conflict manually, and commit the resolved changes.




Subject Faculty Name: Prof. Arijit Dutta